

Insert new clauses Clause 174 through Clause 187 as follows:

174. Introduction to 1.6Tb/s networks

174.1 Overview

174.1.1 Scope

This clause describes the general requirements for 1.6 Terabit Ethernet.

1.6 Terabit Ethernet uses the IEEE 802.3 MAC sublayer operating at a data rate of 1.6 Tb/s, coupled with any IEEE 802.3 1.6TBASE Physical Layer implementation, and is defined for full duplex operation only.

174.1.2 Relationship of 1.6 Terabit Ethernet to the ISO OSI reference model

1.6 Terabit Ethernet couples the IEEE 802.3 MAC to a family of 1.6 Tb/s Physical Layer devices. The relationships among 1.6 Terabit Ethernet, the IEEE 802.3 MAC, and the ISO Open System Interconnection (OSI) reference model are shown in Figure 174–1.

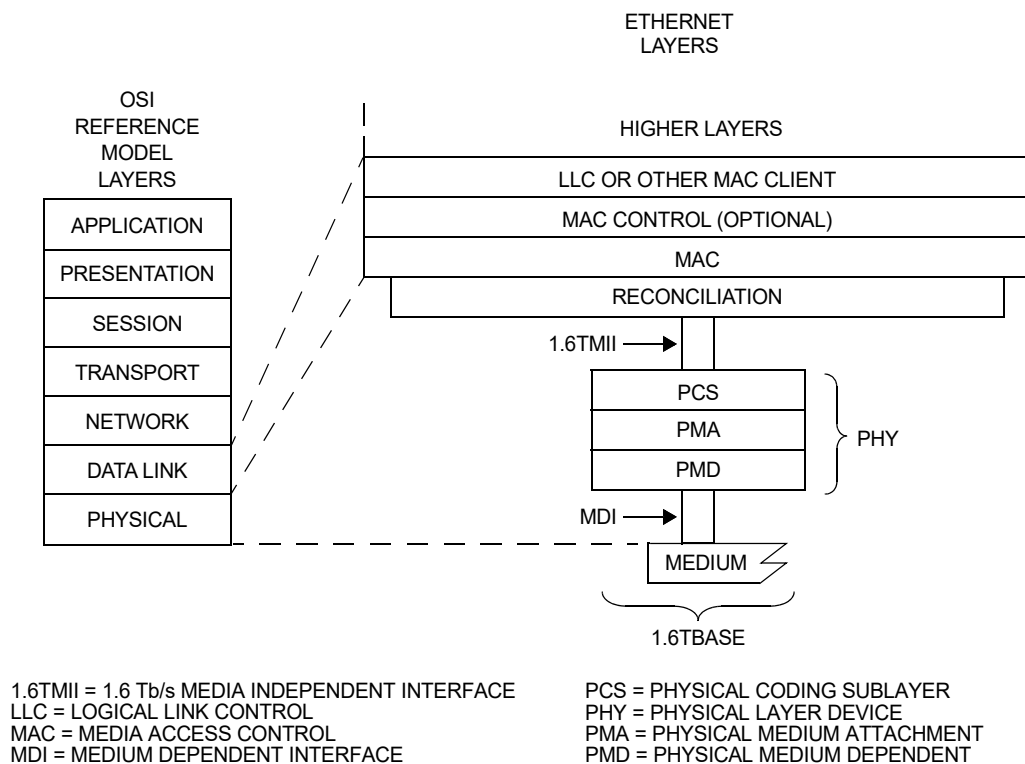


Figure 174–1—Architectural positioning of 1.6 Terabit Ethernet

While this specification defines interfaces in terms of bits, octets, and frames, implementations may choose other data-path widths for implementation convenience, with the following exceptions:

- The management interface, when physically implemented as the MDIO/MDC (see Clause 45, 22.2.2.13, and 22.2.2.14)
- The PMA service interface, when physically implemented as 1.6TAUI-n (see 174.2.5)

— The MDI for a 1.6TBASE PMD (see 174.2.7)

174.1.3 Nomenclature

The nomenclature employed by the 1.6 Tb/s Physical Layer is explained as follows.

The alpha-numeric prefix 1.6TBASE in the port type (e.g., 1.6TBASE-R) represents a family of Physical Layer devices operating at a speed of 1.6 Tb/s.

The term 1.6TBASE-R represents a family of Physical Layer devices using the Physical Coding Sublayer (PCS) defined in Clause 175 for 1.6 Tb/s operation.

Physical Layer devices listed in Table 174–1 are defined for operation at 1.6 Tb/s.

Table 174–1—1.6 Tb/s PHYs

Name	Description
1.6TBASE-KR8	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight lanes of an electrical backplane (see Clause 178)
1.6TBASE-CR8	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight lanes of twinaxial copper cable (see 1.4.559 and Clause 179)
1.6TBASE-DR8	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight single-mode fibers with reach up to at least 500 m (see Clause 180)
1.6TBASE-DR8-2	1.6 Tb/s PHY using 1.6TBASE-R encoding over eight single-mode fibers with reach up to at least 2 km (see Clause 182)

174.1.4 Physical Layer signaling systems

This standard specifies a family of Physical Layer implementations. Table 174–2 and Table 174–3 specify the correlation between PHY types and clauses. Implementations conforming to one or more PHY types meet the requirements of the corresponding clauses.

Table 174–2—PHY type and clause correlation (1.6TBASE-R electrical)

PHY type	Clause ^a									
	73	90	170	171	175	176	120F	176C	178	179
	Auto-Negotiation	Time Synchronization	RS	1.6TMII	1.6TMII Extender	1.6TBASE-R PCS	1.6TBASE-R SM-PMA	1.6TAUI-16 C2C	1.6TAUI-8 C2C	1.6TBASE-KR8 PMD
1.6TBASE-KR8	M	O	M	O	O	M	M	O	O	M
1.6TBASE-CR8	M	O	M	O	O	M	M	O	O	—

^a O = Optional, M = Mandatory.

Table 174–3—PHY type and clause correlation (1.6TBASE-R optical)

PHY type	Clause ^a										
	90	170	171	175	176	177	120F	120G	176C	176D	180
	Time Synchronization	RS	1.6TMII	1.6TMII Extender	1.6TBASE-R PCS	1.6TBASE-R SM-PMA	1.6TBASE-R Inner FEC	1.6TAUI-16 C2C	1.6TAUI-16 C2M	1.6TAUI-8 C2C	1.6TAUI-8 C2M
1.6TBASE-DR8	O	M	O	O	M	M	—	O	O	O	O
1.6TBASE-DR8-2	O	M	O	O	M	M	M	O	O	O	—

^a O = Optional, M = Mandatory.

174.2 Summary of 1.6 Terabit Ethernet architecture

174.2.1 Reconciliation Sublayer (RS) and 1.6 Tb/s Media Independent Interface (1.6TMII)

The 1.6 Tb/s Media Independent Interface (1.6TMII) specified in Clause 170 provides a logical interconnection between the MAC sublayer and Physical Layer devices (PHYs). The 1.6TMII is not intended to be physically instantiated, rather it logically connects layers within a device.

The 1.6TMII supports 1.6 Tb/s operation through its 64-bit-wide transmit and receive data paths. The Reconciliation Sublayer (RS) provides a mapping between the signals provided at the 1.6TMII and the MAC/PLS service definition.

While the 1.6TMII is an optional interface, it is used extensively in this standard as a basis for functional specification and it provides a common service interface for the Physical Coding Sublayer (Clause 175).

174.2.2 1.6TMII Extender and 1.6TMII Extender Sublayer (1.6TXS)

The optional 1.6TMII Extender (Clause 171) may be inserted between the Reconciliation Sublayer and the PHY to transparently extend the reach of the 1.6TMII.

The 1.6TMII Extender Sublayer (1.6TXS) is part of the 1.6TMII Extender. It is equivalent in function to the 1.6TBASE-R PCS in Clause 175 with the exceptions as defined in Clause 171.

174.2.3 Physical Coding Sublayer (PCS)

The PCS performs encoding of data from the 1.6TMII, applies FEC, distributes the data to multiple PCS lanes, and transfers the encoded data to the PMA.

The 1.6TBASE-R PCS is specified in Clause 175.

174.2.4 Physical Medium Attachment (PMA) sublayer

The PMA sublayer provides a medium-independent means to support the use of a range of physical media.

The 1.6TBASE-R SM-PMA is specified in Clause 176.

174.2.5 Attachment unit interface (1.6TAUI-n)

A 1.6 Tb/s Attachment Unit Interface (1.6TAUI-n) provides an electrical interface within a 1.6TBASE-R PHY or 1.6TMII Extender. Two widths of 1.6TAUI-n, 8-lane and 16-lane, are defined for chip-to-chip (C2C) and chip-to-module (C2M) implementations.

The 1.6TAUI-16 C2C is specified in Annex 120F.

The 1.6TAUI-16 C2M is specified in Annex 120G.

The 1.6TAUI-8 C2C is specified in Annex 176C.

The 1.6TAUI-8 C2M is specified in Annex 176D.

Each 1.6TAUI-n is instantiated within a Physical Layer implementation as described in 176B.8.

174.2.6 Inner FEC sublayer

The Inner FEC sublayer provides error correction, in addition to that provided by the 1.6TBASE-R PCS, for the PMD.

For 1.6TBASE-DR8-2 the 1.6TBASE-R Inner FEC is specified in Clause 177.

174.2.7 Physical Medium Dependent (PMD) sublayer

The PMD sublayer is responsible for interfacing to the transmission medium. The PMD is located just above the Medium Dependent Interface (MDI).

The MDI, logically subsumed within each PMD subclause, is the actual medium attachment for the various supported media.

1.6TBASE-R PMDs, including the corresponding MDI and media, are specified in Clause 178, Clause 179, Clause 180, and Clause 182.

174.2.8 Auto-Negotiation (AN)

Auto-Negotiation (AN) provides a device with the capability to detect the abilities (modes of operation) supported by the device at the other end of the link, determine common abilities, and configure for joint operation.

AN used by a 1.6 Tb/s backplane PHY (1.6TBASE-KR8) and the 1.6 Tb/s copper PHY (1.6TBASE-CR8) is specified in Clause 73.

174.2.9 Management interface (MDIO/MDC)

The optional MDIO/MDC management interface provides an interconnection between MDIO Manageable Devices (MMDs) and Station Management (STA) entities.

The MDIO/MDC management interface is specified in Clause 45.

174.2.10 Management

Managed objects, attributes, and actions are defined for all 1.6 Terabit Ethernet components. These items are specified in Clause 30.

174.2.11 Time Synchronization

A 1.6 Tb/s Physical Layer optionally supports time synchronization protocols that require knowledge of packet egress and ingress time. Ethernet support for time synchronization protocols is defined in Clause 90.

If time synchronization is supported, then the following capabilities are provided:

- The 1.6 Tb/s Reconciliation Sublayer (RS) provides a Time Synchronization Service Interface (TSSI) that connects to a TimeSync client (see 90.4).
- The path data delay values of each sublayer in a Physical Layer are reported as status variables, which are optionally mapped to MDIO status registers (see 90.6).

174.2.12 Autonomous path startup (APSU)

Autonomous path startup (APSU) is the coordinated, orderly initialization of all inter-sublayer links (ISLs) and sublayers in a path. Functions to support APSU are defined in Annex 178B.

APSU functions are specified for use by the following PMD and AUI types:

- 1.6TAUI-8 C2C
- 1.6TAUI-8 C2M
- 1.6TBASE-KR8

- 1.6TBASE-CR8
- 1.6TBASE-DR8
- 1.6TBASE-DR8-2

174.2.13 FEC Degrade

The FEC degrade feature provides the ability to detect and indicate a degrade condition at the RS-FEC decoder using FEC degrade detection, and to propagate the FEC degrade indication using FEC degrade signaling. The propagation of FEC degrade indications across PCS and XS is described in 116.6.

FEC degrade detection and signaling specifications for the 1.6TXS and 1.6TBASE-R PCS are summarized in 171.6 and 175.3, respectively.

174.3 Service interface specification method and notation

The service interface specification for the 1.6TBASE-R Physical Layers is as per the definition in 1.2.2. Note that the 1.6TBASE-R inter-sublayer service interfaces use multiple scalar REQUEST and INDICATION primitives, to indicate the transfer of multiple independent streams of data units, as defined in 174.3.1 through 174.3.3.

174.3.1 Inter-sublayer service interface

The inter-sublayer service interface is described in an abstract manner and does not imply any particular implementation. The inter-sublayer service interface primitives are defined as follows:

```
IS_UNITDATA_i.request
IS_UNITDATA_i.indication
IS_SIGNAL.indication
IS_SIGNAL.request
```

The IS_UNITDATA_i.request (where $i = 0$ to $n-1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units to a sublayer from the next higher (closer to the MAC) sublayer.

The IS_UNITDATA_i.indication (where $i = 0$ to $n-1$, and n is the number of streams of data units) primitive is used to define the transfer of multiple streams of data units from a sublayer to the next higher sublayer.

The IS_SIGNAL.indication primitive is used to define the transfer of signal status from a sublayer to the next higher sublayer.

The IS_SIGNAL.request primitive is used to define the transfer of signal status to a sublayer from the next higher sublayer.

174.3.2 Instances of the inter-sublayer service interface

The inter-sublayer service interface may be instantiated between different sublayers, hence a prefix notation is defined to identify a specific instance of an inter-sublayer service interface. The following prefixes are defined:

- PMA:—for primitives issued on the interface between the PMA and the sublayer above called the PMA service interface
- PHY_XS:—for primitives issued on the interface between the PHY 1.6TXS and the sublayer above called the PHY 1.6TXS service interface

- FEC:—for primitives issued on the interface between the Inner FEC and the sublayer above called the FEC service interface
- PMD:—for primitives issued on the interface between the PMD and the sublayer above called the PMD service interface

As an example, the primitives for the PMD service interface are identified as follows:

PMD:IS_UNITDATA_0*i*.request
PMD:IS_UNITDATA_0*i*.indication
PMD:IS_SIGNAL.indication
PMD:IS_SIGNAL.request

Examples of inter-sublayer service interfaces for 1.6TBASE-R with their corresponding instance names are illustrated in Figure 174–2 through Figure 174–4.

Primitives for other instances of inter-sublayer interfaces are represented in a similar manner.

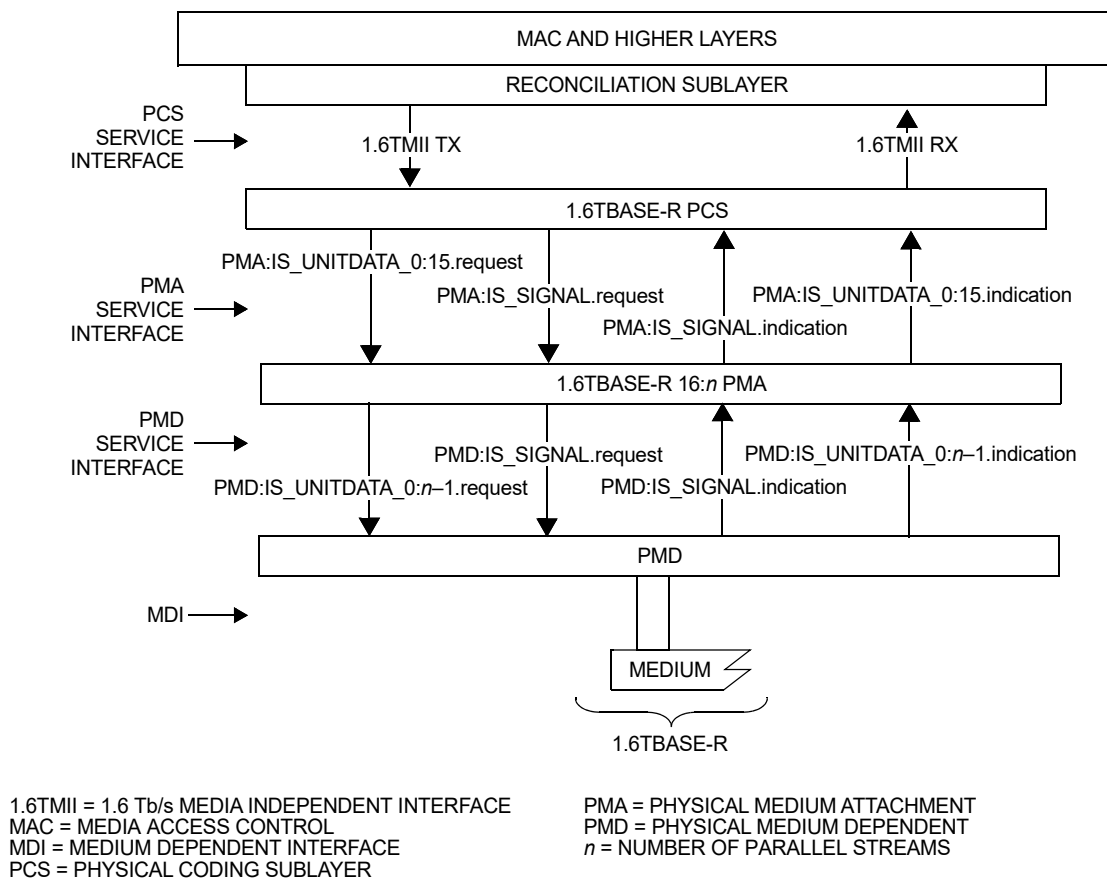
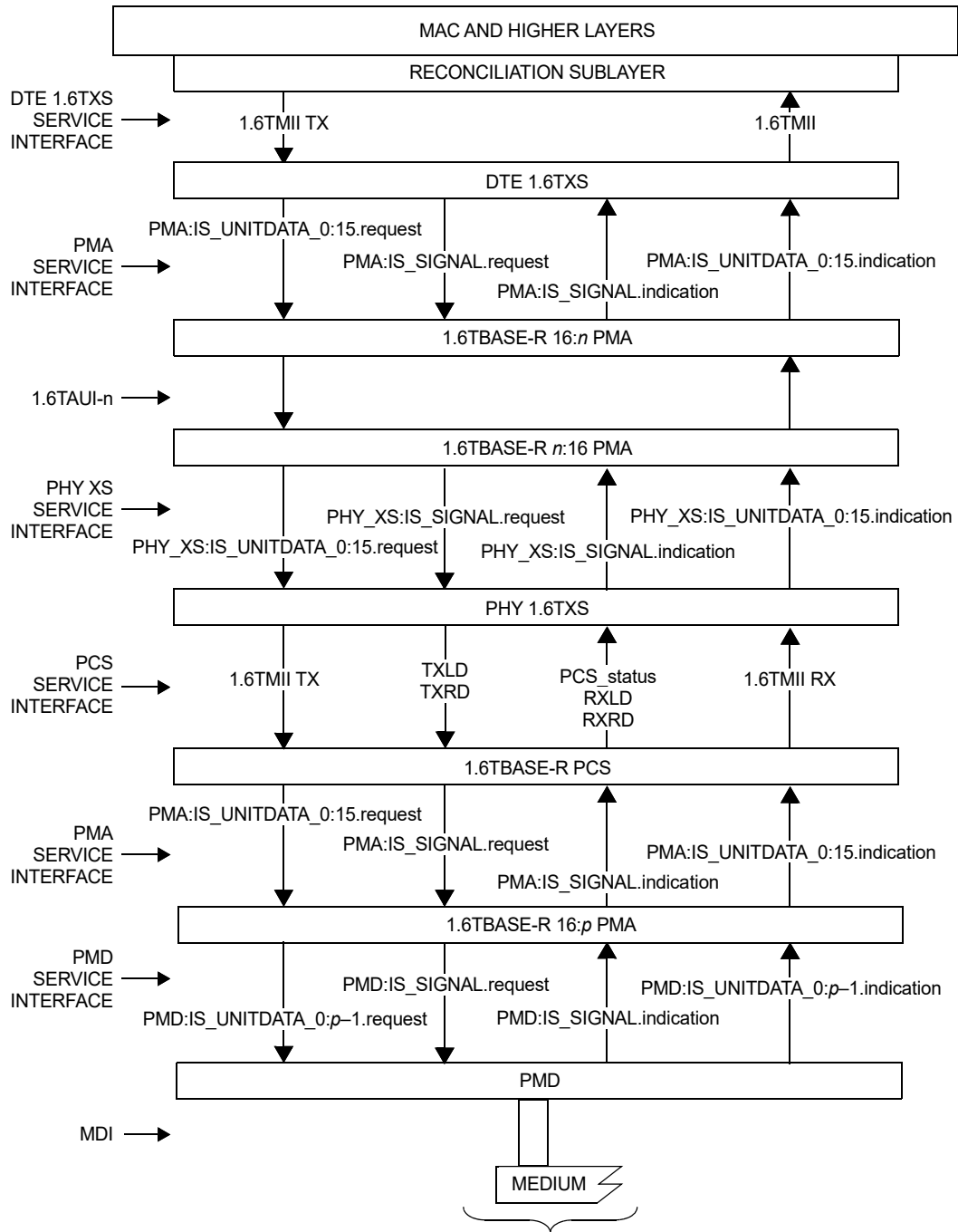


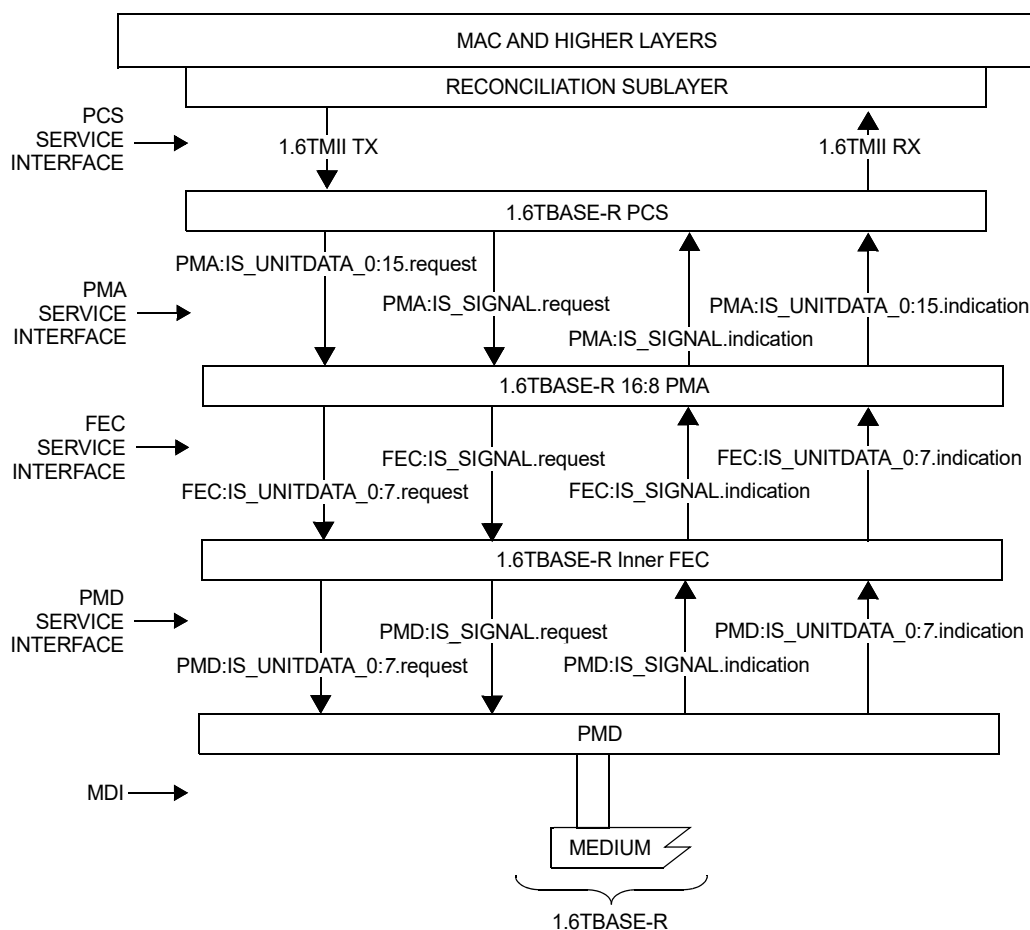
Figure 174–2—Inter-sublayer service interfaces for a 1.6TBASE-R PHY



1.6TAUI-n = 1.6 Tb/s ATTACHMENT UNIT INTERFACE
1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
1.6TXS = 1.6 Tb/s PHY EXTENDER SUBLAYER
DTE = DATA TERMINAL EQUIPMENT
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

1.6TBASE-R
PHY = PHYSICAL LAYER DEVICE
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
XS = EXTENDER SUBLAYER
n = NUMBER OF PARALLEL STREAMS
p = NUMBER OF PARALLEL STREAMS

Figure 174-3—Inter-sublayer service interfaces for a 1.6TBASE-R PHY and 1.6TMII Extender



1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE

PCS = PHYSICAL CODING SUBLAYER
PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT

Figure 174-4—Inter-sublayer service interfaces for a 1.6TBASE-R PHY with 1.6TBASE-R Inner FEC

174.3.3 Semantics of inter-sublayer service interface primitives

The semantics of the inter-sublayer service interface primitives for the 1.6TBASE-R sublayers are described in 116.3.3.

174.4 Delay constraints

Predictable operation of the MAC Control PAUSE operation (Clause 31, Annex 31B) demands that there be an upper bound on the propagation delays through the network. This implies that MAC, MAC Control sublayer, and PHY implementations conform to certain delay maxima, and that network planners and administrators conform to constraints regarding the cable topology and concatenation of devices. Table 174–4 contains the values of maximum sublayer delay (sum of transmit and receive delays at one end of the link). If a PHY contains an Auto-Negotiation sublayer, the delay of the Auto-Negotiation sublayer is included within the delay of the PMD and medium. The delay for a set of sublayers within the same package may be constrained by the sum of constraints for the set of the sublayers.

Table 174–4—Sublayer delay constraints

Sublayer	Maximum			Notes ^a
	(bit time) ^b	(pause_quantum) ^c	(ns)	
1.6T MAC, RS, and MAC Control	393 216	768	245.76	See 170.1.4.
1.6TBASE-R PCS or 1.6TXS ^d	1 280 000	2500	800	See 175.5.
1.6TBASE-R 8:16 or 16:8 PMA	24 576	48	15.36	See 176.8.
1.6TBASE-R 8:8 or 16:16 PMA	24 576	48	15.36	See 176.8.
1.6TBASE-R Inner FEC	138240	270	86.4	See 177.8.
1.6TAUI-8 C2C or C2M component	73 728	144	46.08	Includes allocation of 5 ns for one direction through AUI channel. See 176C.5 and 176D.5.
1.6TBASE-KR8 PMD	118 784	232	74.24	Includes allocation of 14 ns for one direction through the backplane medium. See 178.6.
1.6TBASE-CR8 PMD	118 784	232	74.24	Includes allocation of 14 ns for one direction through the cable medium. See 179.6.
1.6TBASE-DR8 PMD	118 784	232	74.24	Includes 2 m of fiber. See 180.4.1.
1.6TBASE-DR8-2 PMD	118 784	232	74.24	Includes 2 m of fiber. See 182.4.1.

^a Should there be a discrepancy between this table and the delay requirements of the relevant sublayer clause, the sublayer clause prevails.

^b For 1.6TBASE, 1 bit time is equal to 0.625 ps. (See 1.4.215 for the definition of bit time.)

^c For 1.6TBASE, 1 pause_quantum is equal to 320 ps. (See 1.4.459 for the definition of pause_quantum.)

^d If an implementation includes the 1.6TMII Extender, the delay associated with the 1.6TMII Extender includes two 1.6TXS sublayers.

The physical medium interconnecting two optical PHYs introduces additional delay in a link.

See 80.4 for the calculation of bit time per meter of fiber or electrical cable.

See 31B.3.7 for PAUSE reaction timing constraints for stations at operating speeds of 1.6 Tb/s.

174.5 Skew constraints

Skew (or relative delay) can be introduced between PCS lanes by both active and passive elements of a 1.6TBASE-R link. Skew is defined as the difference in time between the start of the alignment marker for the earliest and latest arriving PCS lanes. The PCS deskew function (see 175.2.5.1) compensates for all lane-to-lane Skew observed at the receiver. The Skew between the PCS lanes is kept within limits as shown in Table 174–5 so that the transmitted information on the PCS lanes can be reassembled by the receive PCS.

Skew Variation may be introduced due to variations in electrical, thermal or environmental characteristics. Skew Variation is defined as the change in Skew between any PCS lane and any other PCS lane over the entire time that the link is in operation. From the time the link is brought up, Skew Variation is limited so that each PCS lane always traverses the same lane between any pair of adjacent sublayers while the link remains in operation.

The maximum Skew and Skew Variation at physically instantiated interfaces is specified at Skew points SP1, SP2, and SP3 for the transmit direction and SP4, SP5, and SP6 for the receive direction as illustrated in Figure 174–5 for a PHY with one 1.6TAUI-n and Figure 174–6 for a PHY with two 1.6TAUI-n.

NOTE—No physically instantiated interfaces are specified at SP2 and SP5 (PMD service interface) for 1.6TBASE-R PHY types.

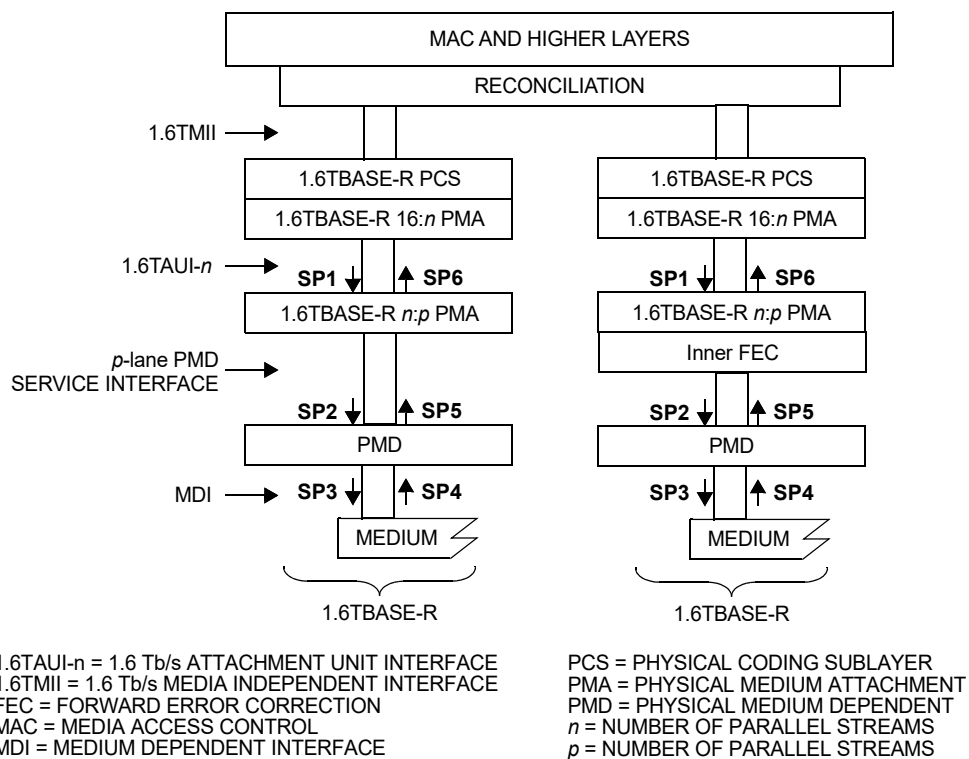
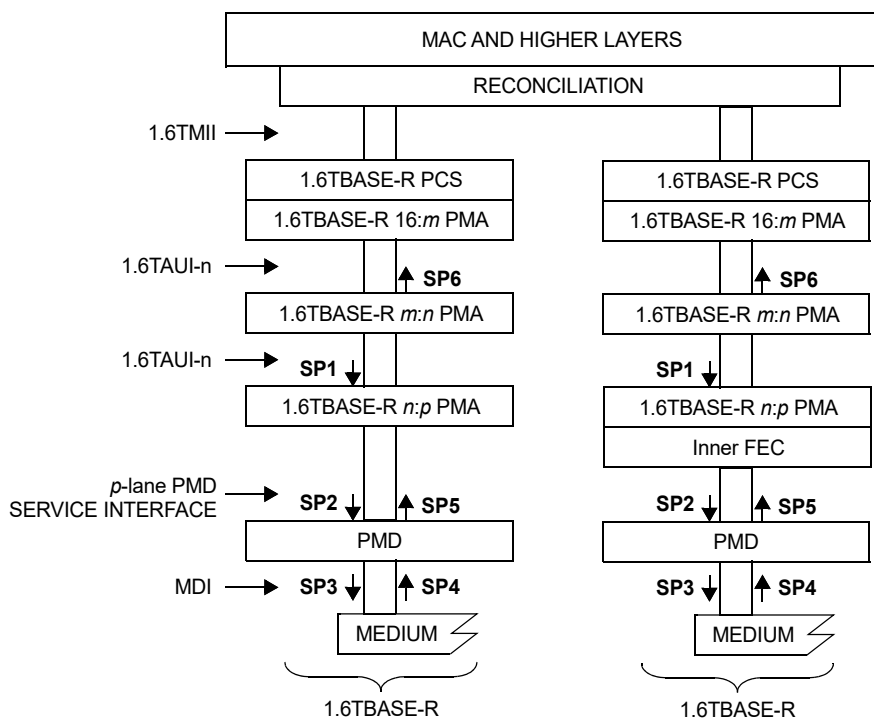


Figure 174–5—1.6TBASE-R Skew points for a PHY with one 1.6TAUI-n



1.6TAUI-n = 1.6 Tb/s ATTACHMENT UNIT INTERFACE
1.6TMII = 1.6 Tb/s MEDIA INDEPENDENT INTERFACE
FEC = FORWARD ERROR CORRECTION
MAC = MEDIA ACCESS CONTROL
MDI = MEDIUM DEPENDENT INTERFACE
PCS = PHYSICAL CODING SUBLAYER

PMA = PHYSICAL MEDIUM ATTACHMENT
PMD = PHYSICAL MEDIUM DEPENDENT
 m = NUMBER OF PARALLEL STREAMS
 n = NUMBER OF PARALLEL STREAMS
 p = NUMBER OF PARALLEL STREAMS

Figure 174-6—1.6TBASE-R Skew points for a PHY with two 1.6TAUI-n

In the transmit direction, the Skew points are defined in the following locations:

- SP1 on the 1.6TAUI-n interface closest to the PMD at the input of the PMA
- SP2 on the PMD service interface, at the input of the PMD
- SP3 at the output of the PMD, at the MDI

In the receive direction, the Skew points are defined in the following locations:

- SP4 at the MDI, at the input of the PMD
- SP5 on the PMD service interface, at the output of the PMD
- SP6 on the 1.6TAUI-n interface closest to the PCS or DTE 1.6TXS at the output of the PMA

The allowable limits for Skew are shown in Table 174-5 and the allowable limits for Skew Variation are shown in Table 174-6.

The Skew and Skew Variation requirements for the PCS, PMA and PMD sublayers are specified in the respective clauses as noted in Table 174-5 and Table 174-6.

Table 174–5—Summary of Skew constraints

Skew points	Maximum Skew		Notes ^a
	(ns) ^b	for 1.6TBASE-R PCS lane (bit) ^c	
SP1	16	≈ 1 700	See 176.9
SP2	N/A	N/A	
SP3	36	≈ 3 825	See 178.7, 179.7, 180.4.2, 182.4.2
SP4	116	≈ 12 325	See 178.7, 179.7, 180.4.2, 182.4.2
SP5	N/A	N/A	
SP6	145	≈ 15 406	See 176.9
At PCS receive	152	≈ 16 150	See 175.2.5.1

^a Should there be a discrepancy between this table and the Skew requirements of the relevant sublayer clause, the sublayer clause prevails.

^b The Skew limit includes 1 ns allowance for PCB traces that are associated with the Skew points.

^c The symbol ≈ indicates approximate equivalent of maximum Skew in bits based on 1 bit time equals 9.411765 ps at PCS lane signaling rate of 106.25 Gb/s.

Table 174–6—Summary of Skew Variation constraints

Skew points	Maximum Skew Variation					Notes ^a
	(ns)	for 106.25 Gb/s PCS lane (bit) ^b	for 53.125 GBd PMD lane (UI) ^c	for 106.25 GBd PMD lane (UI) ^d	for 113.4375 GBd PMD lane (UI) ^e	
SP1	0.2	N/A	≈ 11	≈ 21	N/A	See 176.9
SP2	N/A	N/A	N/A	N/A	N/A	
SP3	0.6	N/A	N/A	≈ 64	≈ 68	See 178.7, 179.7, 180.4.2, 182.4.2
SP4	3.4	N/A	N/A	≈ 361	≈ 386	See 178.7, 179.7, 180.4.2, 182.4.2
SP5	N/A	N/A	N/A	N/A	N/A	
SP6	3.8	N/A	≈ 202	≈ 404	N/A	See 176.9
At PCS receive	4	≈ 425	N/A	N/A	N/A	See 175.2.5.1, 176.9

^a Should there be a discrepancy between this table and the Skew Variation requirements of the relevant sublayer clause, the sublayer clause prevails.

^b The symbol ≈ indicates approximate equivalent of maximum Skew Variation in bits based on 1 bit time equals 9.411765 ps at PCS lane signaling rate of 106.25 Gb/s.

^c The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 18.82353 ps at PMD lane signaling rate of 53.125 GBd.

^d The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 9.411765 ps at PMD lane signaling rate of 106.25 GBd.

^e The symbol ≈ indicates approximate equivalent of maximum Skew Variation in UI based on 1 UI equals 8.81543 ps at PMD lane signaling rate of 113.4375 GBd.

174.6 Protocol implementation conformance statement (PICS) proforma

The supplier of a protocol implementation that is claimed to conform to any part of IEEE Std 802.3, Clause 175 through Clause 180 and Clause 182, and related annexes demonstrates compliance by completing a protocol implementation conformance statement (PICS) proforma.

A completed PICS proforma is the PICS for the implementation in question. The PICS is a statement of which capabilities and options of the protocol have been implemented. A PICS is included at the end of each clause as appropriate. Each of the 1.6 Terabit Ethernet PICS conforms to the notation and conventions used in 21.6.